Freight Rate Prediction

Validation report | Arafat Rahman

Data: 48,000 labeled loads from January through October 2025.
The task is to predict 12,000 November/December loads.
The target is posted_rate, in dollars.

Data checks: 300 missing weights and 374 missing market index values.
Numeric fields use median imputation within each training split.
There are 0 duplicate predictor rows and 736 new routes in the prediction data.
Unknown categories are handled by the encoders.

Split: train through July to evaluate August, then train through August
to evaluate September. Choose settings by mean development MAE.
Keep October for testing. IDs are excluded; preprocessing is fitted
on each training portion. Refit on all labeled rows for submission.

Models compared: median baseline, Ridge and three histogram gradient
boosting settings. Selected settings:
Full inputs: hgb_3. Six December fields: hgb_1.

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October test metrics (USD for MAE/RMSE):
mode     MAE    RMSE    R2  MAPE_pct
full 196.657 672.675 0.806     9.374
core 169.050 661.170 0.813     8.827
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October MAE intervals: resample whole days 1,000 times and use the
2.5th and 97.5th percentiles. These describe variation across October
days, rather than uncertainty in each future load prediction.

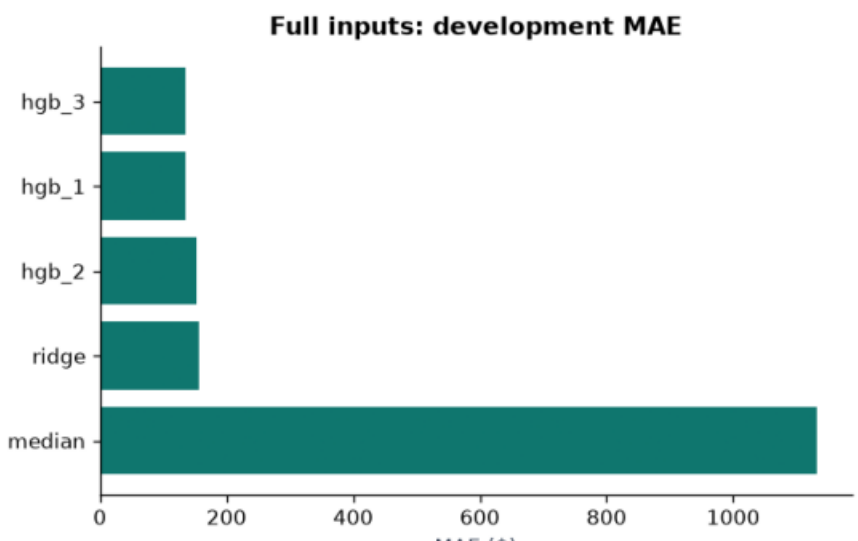
mode	MAE_day_bootstrap_low	MAE_day_bootstrap_high
full	180.515	212.460
core	152.323	186.377

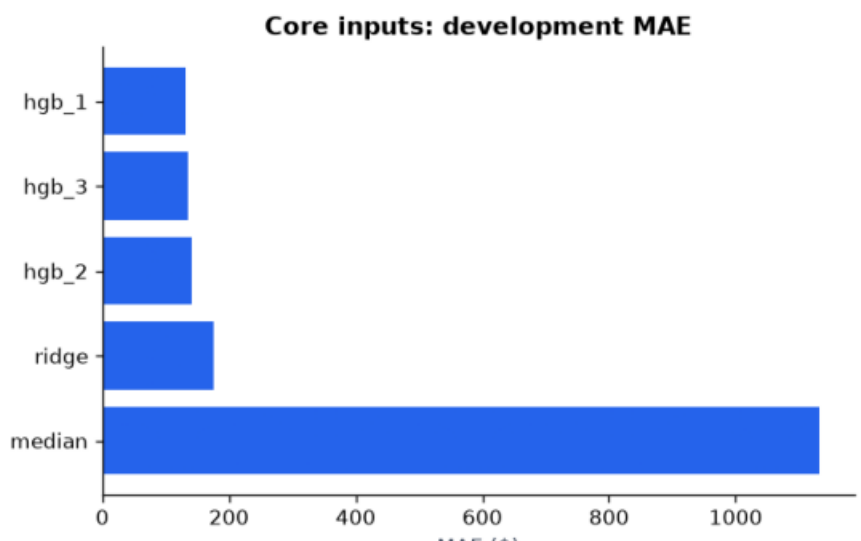
December chart: pickup, delivery, distance, equipment and weight stay
fixed; only the date changes. A separate model uses these six fields
because market signals and coordinates are absent from that file.

Limits: November/December labels are unavailable. The extra two-month
check overlaps September development data. Ten months of training
partly cover annual seasonality; tree models also have limited ability
to extend market trends. The December values are a forecast.

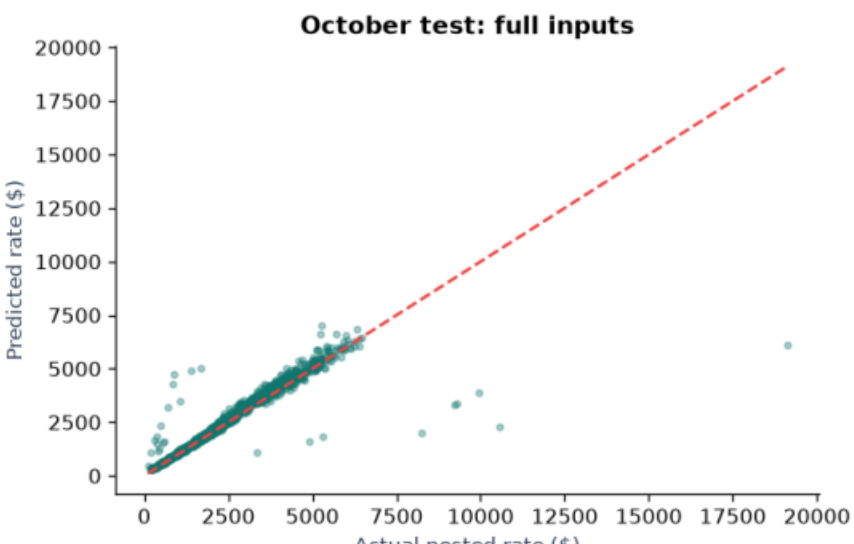
quote_signal is supplied for validation and is used as an input.
Its availability before the quoted rate would need checking in production.
The supplied scorer checks both CSVs and creates the December chart;
it does not calculate the hidden-label score.

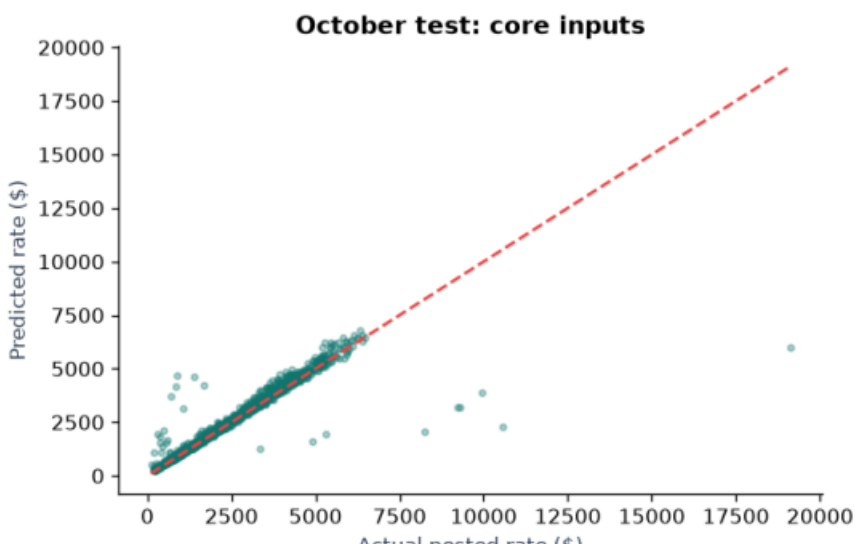
Model comparison





October diagnostics





December 2025 fixed-lane forecast

